

CINDER v1.1.2 Ballew study — interpretation memo

Interpretation memo: CINDER v1.1.2 vs Ballew benchmark

Working interpretation to preserve:

- 1) Closed-loop shaft-speed agreement is reasonably good and stable.
- 2) Force replay collapses, which indicates that the mapping from clamp force -> shift/contact response differs materially between the models.
- 3) Therefore this bridge should be presented as a macroscopic controlled-response comparison rather than internal validation.

Closed-loop aggregate metrics:

Primary RPM RMSE	=	110.322 rpm
Primary RPM RMSE %	=	4.415 %
Secondary RPM RMSE	=	32.580 rpm
Secondary RPM RMSE %	=	2.714 %
Shaft-speed-ratio RMSE	=	0.10020
Shaft-speed-ratio RMSE %	=	4.806 %
Primary clamp RMSE	=	1196.249 N
Primary clamp RMSE %	=	46.661 %

Force-replay aggregate metrics:

Primary RPM RMSE %	=	77.886 %
Secondary RPM RMSE %	=	1.090 %
Shaft-speed-ratio RMSE %	=	78.305 %

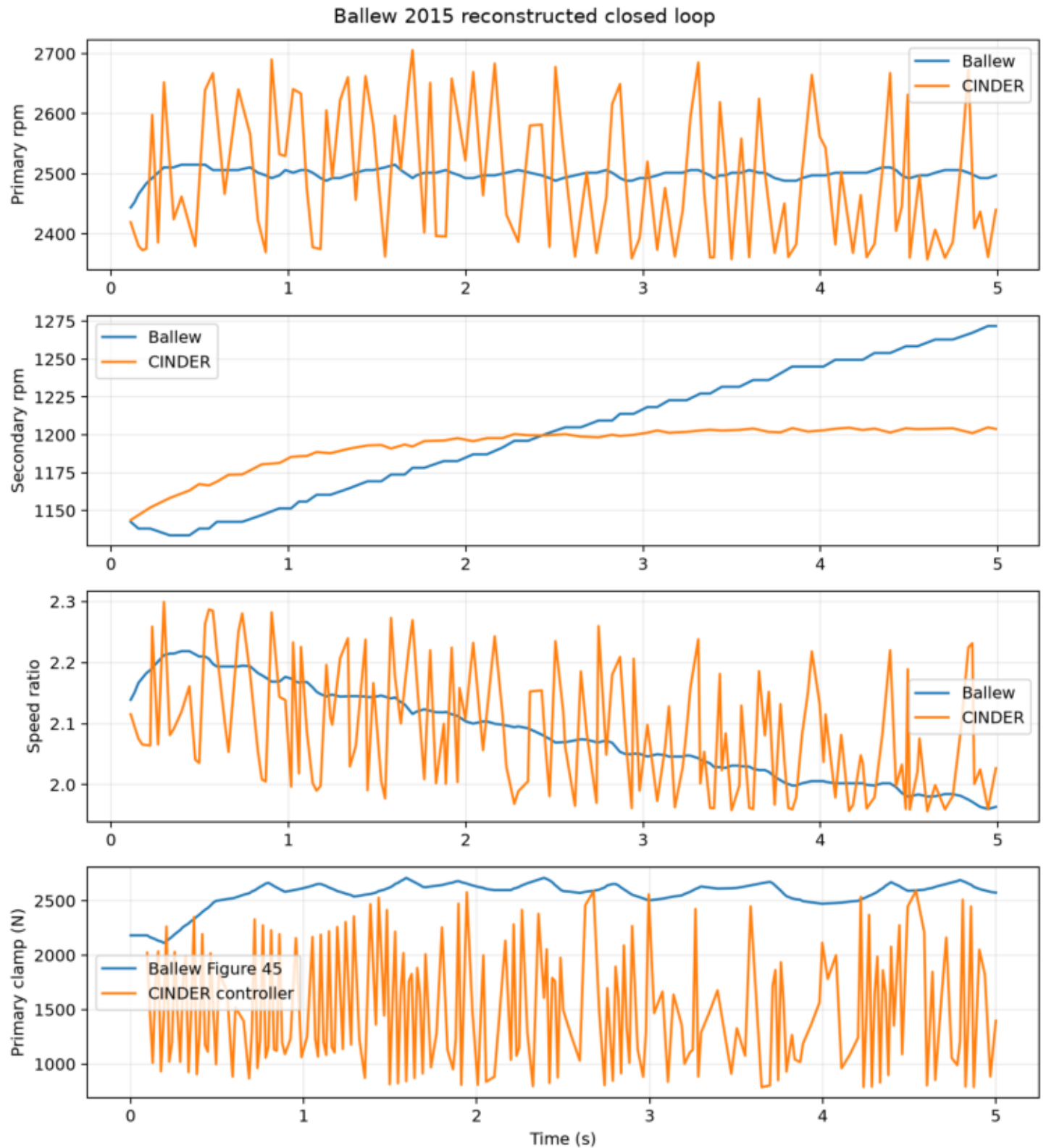
Internal caution:

CINDER's geometric ratio evolves very differently from the shaft-speed ratio in this closed-loop bridge. That makes the study useful, but it should not be sold as validation of internal shift/contact behavior.

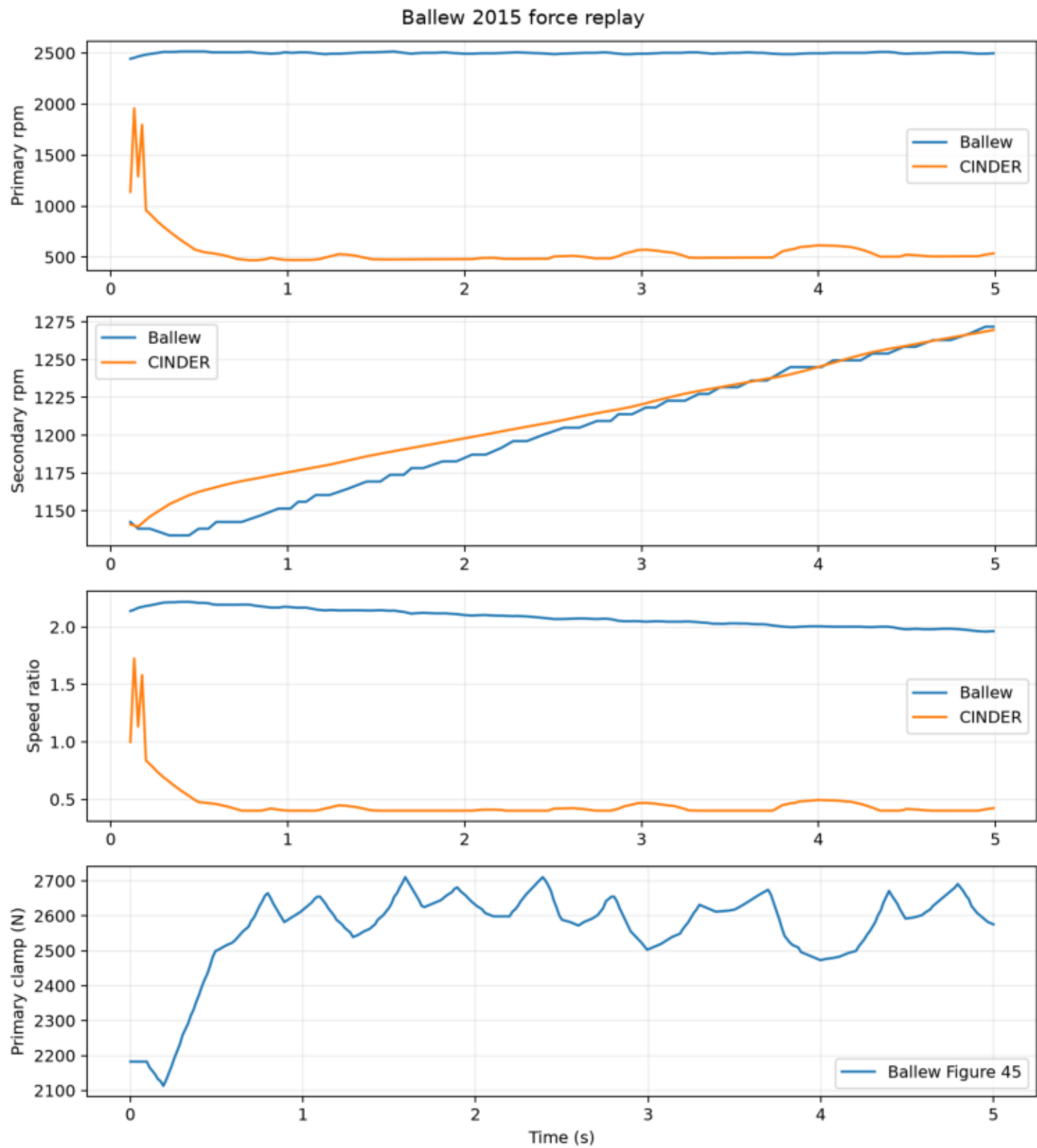
Status:

Promising enough to keep and reuse, with careful wording in the final writeup.

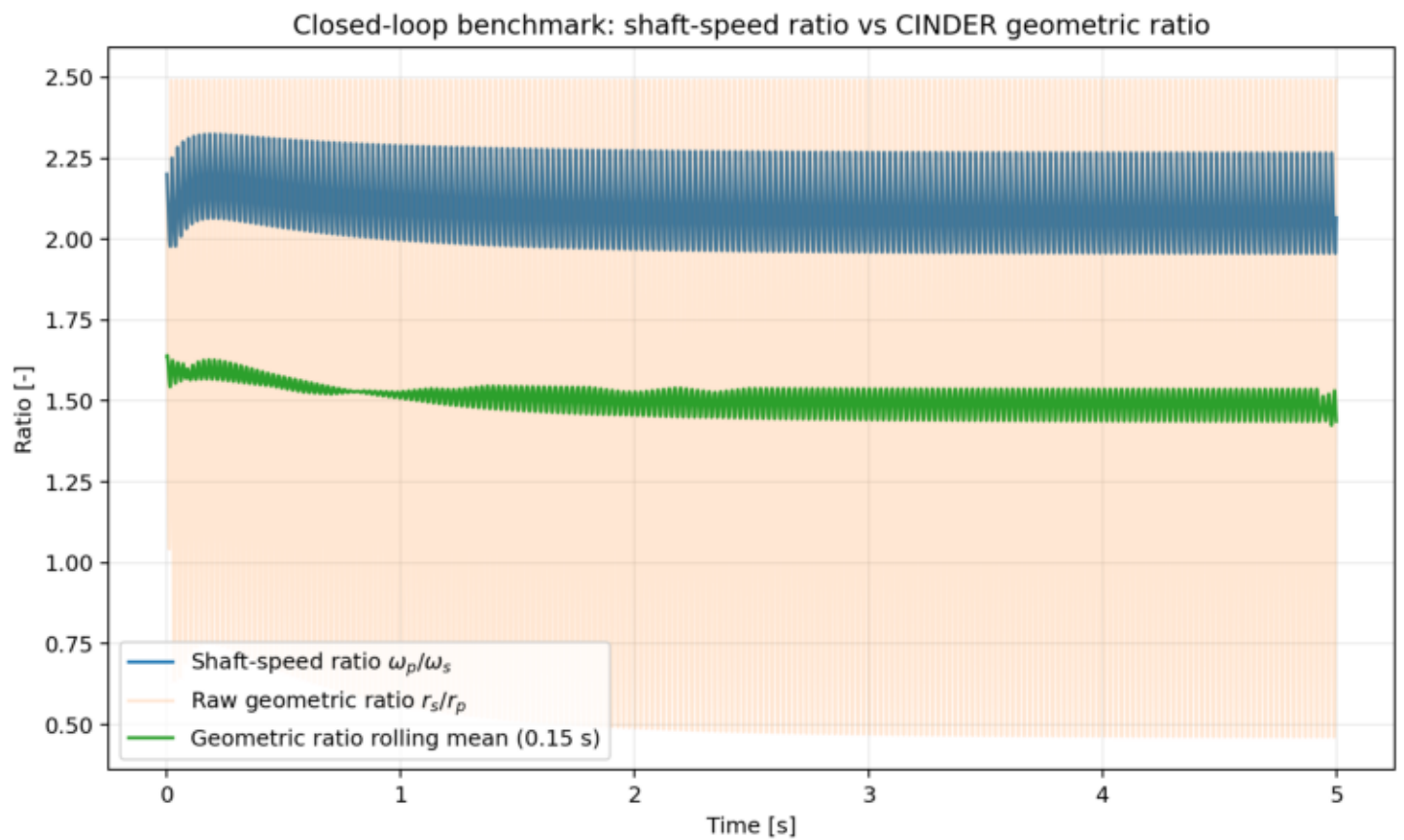
Closed-loop comparison



Force-replay comparison

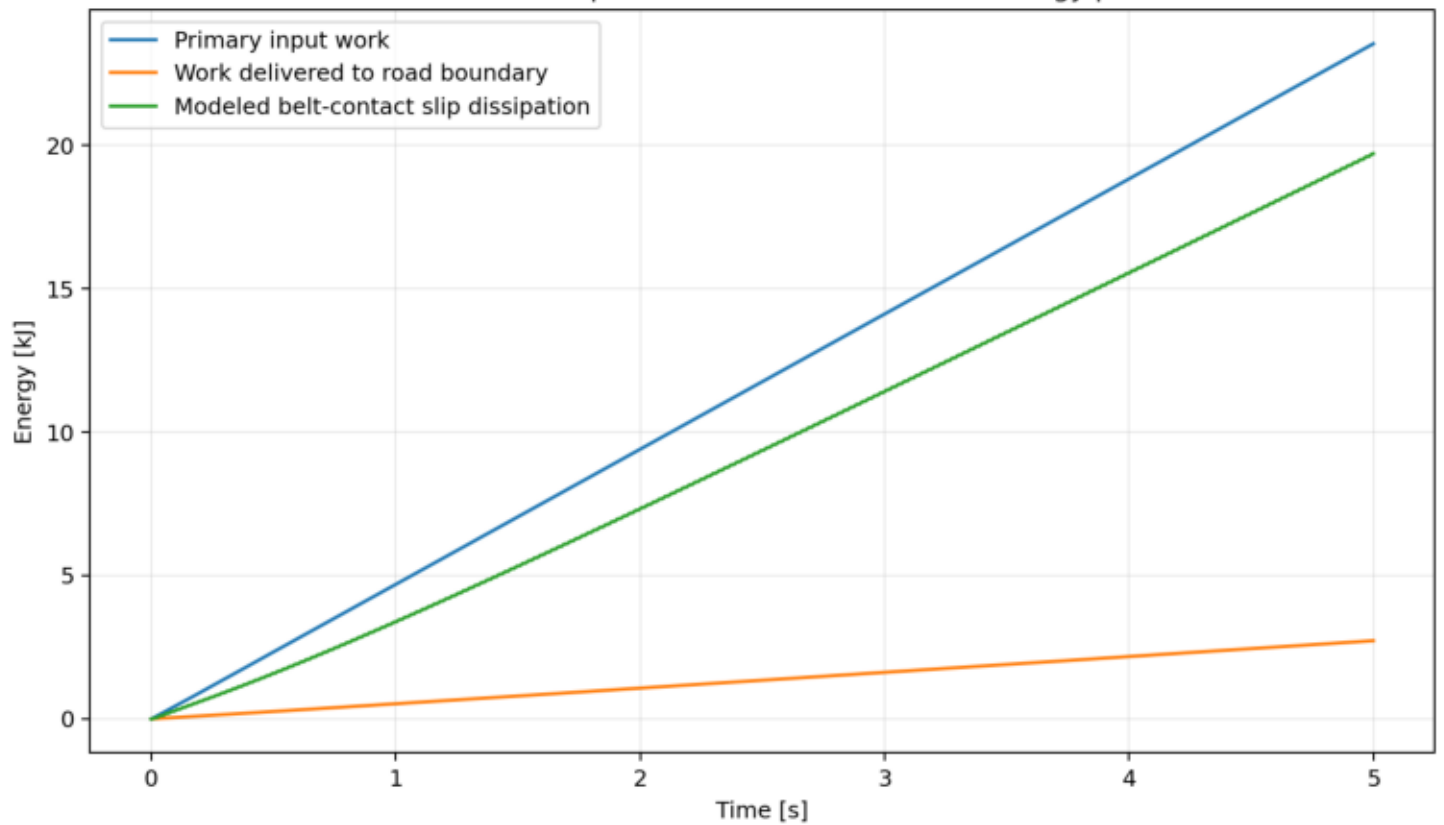


Closed-loop shaft-speed ratio vs geometric ratio



Closed-loop cumulative energy paths

Ballew closed-loop reconstruction: cumulative energy paths



Closed-loop comparison (full scale, supporting view)

Ballew 2015 reconstructed closed loop — full scale

